CNC Spindle 200 Maintenance Manual

SP-200 belt-driven machining spindle

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Kestrel Spindle Systems (fictional)

Read this manual before installing, operating or servicing the equipment. Keep it at the machine.

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1 Safety

1.1 Intended use

The CNC Spindle 200 (SP-200) is a belt-driven machining spindle for drilling, boring and milling of aluminium and steel housings in automated cells. Use it only within the speed, power and load limits in section 2.3, with the original drive belt, guards and sensors fitted. Any other use is not intended and voids the warranty.

The spindle is part of a machine. The machine builder is responsible for the guards, the emergency stop circuit and the integration into the line control. Follow this manual together with the line documentation and the plant's own safety rules.

1.2 Qualified personnel

Only trained and authorised people may operate or service the spindle. The table shows the minimum qualification for each task.

Task	Who	Qualification
Daily checks (7.1)	Operator	Line training, spindle module
Weekly checks (7.2)	Operator or technician	Line training
Re-greasing (5.4)	Maintenance technician	Mechanical trade
Semi-annual service (7.3)	Maintenance technician	Mechanical trade, certificate spindle-200
Bearing replacement (7.5)	Certified technician	Certificate spindle-200
Belt alignment (7.6)	Technician	Laser alignment training
Electrical work	Electrician	Qualified electrician, plant switching rules

1.3 Safety symbols

Symbol	Meaning
Warning	Danger to life or risk of serious injury.
Caution	Risk of minor injury or damage to the equipment.
Note	Useful information, no hazard.
Hot surface	Bearing housing and motor can exceed 60 °C.
Rotating parts	Never reach into the working area while the spindle turns.
Stored energy	Drawbar spring and sealing air remain loaded after switch-off.

1.4 Lockout and tagout

Before any work inside the guard, on the belt drive or on the bearings, isolate all energy sources and prove that the machine is safe.

1. Stop the spindle with the line control and wait until it has stopped (at least 20 s after 12,000 rpm).
2. Switch off and lock the main isolator of the spindle drive. Attach your personal lock and tag.
3. Close and lock the sealing-air valve and vent the line.
4. Close the coolant supply valve and switch off the chiller.
5. Release the drawbar spring with the release tool if the tool holder will be removed.

6 Try to start the spindle from the panel to prove the isolation. Return the selector to OFF.

7 Only now open the guard or the belt cover.

Warning	Never rely on the drive's safe torque off alone for work on the belt drive or bearings. Always lock the main isolator.

1.5 Personal protective equipment

Task	Required PPE
Operation	Safety glasses, safety shoes, hearing protection above 85 dB(A)
Cleaning and chip removal	Cut-resistant gloves, safety glasses
Re-greasing	Nitrile gloves, safety glasses
Bearing replacement	Heat-resistant gloves for the induction heater, safety glasses
Coolant work	Nitrile gloves, goggles, apron

1.6 Residual risks

Hazard	Situation	Protection
Ejected tool or chips	Tool breakage at high speed	Guard closed, tool checked before use
Burns	Touching housing after operation	Wait 30 min or wear gloves
Crushing	Lifting the spindle unit (48 kg)	Use the lifting eye and a hoist
Hearing damage	Long work near running spindle	Hearing protection
Skin irritation	Coolant and grease	Gloves, skin care
Unexpected start	Remote start by line control	Lockout and tagout (1.4)

1.7 Emergency stop and fire

The emergency stop brakes the spindle to standstill within 3 s and switches off the drive. In case of fire, stop the line, switch off the main isolator and use a CO2 extinguisher. Do not use water on the drive cabinet.

2 Description

2.1 Overview

The SP-200 consists of a cartridge spindle with an HSK-A63 tool interface, an 11 kW asynchronous motor and an HTD 8M toothed-belt drive with a 1:1.6 ratio. The shaft runs on two angular contact ball bearings type 7008 at the front (tool end, back-to-back) and one deep groove ball bearing 6205-2RS at the rear. The rear bearing is the free (floating) bearing and carries the belt pull, which makes it the bearing most exposed to wear.

Sensors for vibration, bearing temperature, motor current, power and speed are fitted at the factory and connected to the line's edge gateway. Chapter 6 describes the signals and how to read them.

2.2 Main components

Item	Description	Part number
Spindle cartridge	HSK-A63, max. 15,000 rpm	SP-200-CART
Front bearings	2 x angular contact 7008, back-to-back	SKF-7008
Rear bearing	Deep groove ball bearing 6205-2RS	SKF-6205
Drive belt	HTD 8M toothed belt, 30 mm	BLT-HTD-8M
Motor pulley	HTD 8M, 32 teeth	PUL-8M-32
Spindle pulley	HTD 8M, 20 teeth	PUL-8M-20
Drive motor	11 kW, 4-pole, 400 V	MTR-SP200-11
Drawbar unit	Spring package, 18 kN clamping force	DRB-63
Rotary union	Through-spindle coolant	RU-63
Vibration sensor	Accelerometer, 100 mV/g	VIB-SENS-100
Bearing temperature	PT100 at rear bearing	PT100-PROBE
Encoder	1,024 lines, sine/cosine	ENC-1024
Shaft seal	Labyrinth plus sealing air	SEAL-LAB-40
Coolant filter	Spindle cooling circuit, 10 µm	FLT-SPC-10

2.3 Technical data

Item	Value
Rated power (S1)	11 kW
Rated speed	12,000 rpm
Maximum speed	15,000 rpm
Rated torque at spindle	8.8 Nm
Belt ratio	1 : 1.6
Motor speed at 12,000 rpm spindle	7,500 rpm
Rated current at 400 V	21 A
Tool interface	HSK-A63
Radial run-out at taper	max. 2 µm
Weight of cartridge	48 kg
Normal vibration RMS at 12,000 rpm	1.8–2.4 mm/s
Normal vibration peak	5–6.5 mm/s
Normal kurtosis	2.7–3.3
Normal rear bearing temperature	55–60 °C
Bearing temperature warning / trip	75 °C / 85 °C
Normal power at full load	10.5–11.5 kW
Vibration alarm limit	4.5 mm/s RMS for 30 min
Cooling	Water-glycol, 3–5 l/min, 20–25 °C
Sealing air	4–6 bar, oil-free
Noise	< 78 dB(A) at 1 m
Ambient temperature	10–40 °C
Protection class	IP54 (spindle), IP55 (motor)
Service interval	4,000 operating hours or 6 months
Re-greasing interval (rear bearing)	2,000 operating hours

2.4 Bearing arrangement

The two front bearings are preloaded by spacers and fix the shaft axially. The rear bearing can slide in its housing to take up thermal growth. Because the belt pulls on the rear end of the shaft, the rear bearing carries most of the radial load. Its condition decides how long the spindle runs between overhauls.

Position	Type	Preload / fit	Lubrication
Front (2x)	7008 angular contact	Light preload, 150 N	Grease for life
Rear	6205-2RS deep groove	Floating, k5 shaft fit	Re-grease every 2,000 h
Motor	6206-2Z	Standard	Grease for life

2.5 Sensors and signal tags

Signals reach the plant systems through the edge gateway. Tag names follow the pattern L{line}.ST{station}.SPDL{n}.{SIGNAL}, for example L3.ST2.SPDL2.VIB_RMS for spindle 2 at line 3, station 2.

Tag suffix	Signal	Unit	Range
VIB_RMS	Vibration RMS (10 Hz–1 kHz)	mm/s	0–20
VIB_PK	Vibration peak	mm/s	0–50
VIB_KURT	Kurtosis of vibration	–	1–20
BRG_T	Rear bearing temperature	°C	0–120
MTR_I	Motor current	A	0–60
PWR	Electrical power	kW	0–30
RPM	Spindle speed	rpm	0–15,000

2.6 Nameplate

Field	Example
Type	SP-200
Serial number	SP2-xxxxx
Year	2021
Max. speed	15,000 rpm
Power	11 kW S1
Weight	48 kg
Manual	MAN-SPINDLE-200

2.7 Dimensions and interfaces

Item	Value
Cartridge diameter	170 mm
Cartridge length	412 mm
Pulley centre distance	260 mm
Coolant connection	G 3/8
Sealing air connection	G 1/8
Rotary union	G 1/4, max. 70 bar
Mounting	6 x M10 on 150 mm pitch circle

2.8 Input and output signals

Address	Type	Signal	Description
I0.0	DI	SPDL_READY	Drive ready
I0.1	DI	SPDL_FAULT	Drive fault
I0.2	DI	SPDL_AT_SPEED	Speed reached
I0.3	DI	SPDL_ZERO	Zero speed
I0.4	DI	TOOL_CLAMPED	Tool clamped
I0.5	DI	TOOL_UNCLAMPED	Tool unclamped
I0.6	DI	DRB_PRESS_OK	Drawbar pressure OK
I0.7	DI	AIR_OK	Sealing air above 4 bar
I1.0	DI	COOL_FLOW_OK	Cooling flow OK
I1.1	DI	CHILLER_OK	Chiller ready
I1.2	DI	BELT_COVER	Belt cover closed
I1.3	DI	GUARD_CLOSED	Guard closed
I1.4	DI	ESTOP_OK	Emergency stop circuit closed
I1.5	DI	LUBE_DUE	Re-greasing due
I1.6	DI	MTR_TEMP_OK	Motor thermistor OK
Q1.7	DO	SPDL_ENABLE	Enable drive
Q2.0	DO	SPDL_START	Start spindle
Q2.1	DO	SPDL_DIR	Direction CCW
Q2.2	DO	TOOL_RELEASE	Release tool
Q2.3	DO	AIR_ON	Sealing air valve
Q2.4	DO	COOL_ON	Cooling pump
Q2.5	DO	THRU_COOL_ON	Through-spindle coolant
Q2.6	DO	BLOW_ON	Taper cleaning air
Q2.7	DO	WARMUP_ACTIVE	Warm-up program running
Q3.0	DO	SPEED_LIMIT	Reduced-speed mode active
IW114	AI	SPEED_ACT	Actual speed 0-15,000 rpm
IW116	AI	SPEED_SET	Set speed 0-15,000 rpm
IW118	AI	LOAD_PCT	Spindle load 0-200 %
IW120	AI	CURRENT	Motor current 0-60 A
IW122	AI	POWER	Power 0-30 kW
IW124	AI	BRG_TEMP	Rear bearing 0-120 degC
IW126	AI	MTR_TEMP	Motor winding 0-180 degC
IW128	AI	COOL_TEMP	Coolant supply 0-60 degC
IW130	AI	COOL_FLOW	Coolant flow 0-10 l/min
IW132	AI	AIR_PRESS	Sealing air 0-10 bar
IW134	AI	VIB_RMS	Vibration RMS 0-20 mm/s
IW136	AI	VIB_PK	Vibration peak 0-50 mm/s

Address	Type	Signal	Description
IW138	AI	VIB_KURT	Kurtosis 1-20
IW140	AI	DRB_FORCE	Clamping force 0-25 kN
IW142	AI	BELT_SLIP	Speed difference 0-10 %

2.9 Declaration of conformity

The manufacturer declares that the SP-200 spindle unit, as a partly completed machine, meets the relevant essential requirements of the Machinery Directive and may only be put into service once the machine into which it is built has been declared in conformity. Applied standards include EN ISO 12100 (risk assessment), EN 60204-1 (electrical equipment) and ISO 10816-3 (vibration evaluation). This declaration is part of a fictional demo manual.

3 Transport and installation

3.1 Transport and lifting

Transport the cartridge only in its original case with the transport lock fitted. Lift it with the lifting eye screwed fully into the rear thread; never lift by the pulley or the cables.

- Keep the spindle horizontal during transport.
- Avoid shocks; a drop of a few centimetres can dent the raceways (false brinelling).
- Store and transport between 5 and 40 °C, dry.

3.2 Storage

Spindles in storage must be turned by hand ten revolutions every month so that the grease spreads and the balls do not mark the raceways. After more than 12 months of storage, run the full warm-up program (4.1) twice before production.

3.3 Mounting

1. Clean the mounting surface; flatness must be within 0.01 mm.
2. Fit the cartridge and tighten the six M10 screws crosswise to 49 Nm.
3. Fit the spindle pulley; tighten the clamping set to 12 Nm.
4. Fit the belt and set the tension (3.4).
5. Connect cooling, sealing air and the rotary union.
6. Connect the sensors and the encoder; check every signal on the HMI.
7. Run the commissioning checklist (3.6) and record the vibration baseline (3.7).

3.4 Belt and pulley alignment

Poor belt alignment loads the rear bearing and is a common cause of early bearing failure. Align the pulleys with the laser tool and set the belt tension with the frequency meter.

Check	Limit	Tool
Parallel offset of pulleys	max. 0.10 mm	Laser pulley tool
Angular misalignment	max. 0.25°	Laser pulley tool
Belt tension (new belt)	142 Hz	Frequency meter
Belt tension (after run-in)	128–136 Hz	Frequency meter

3.5 Electrical connection

Connection	Specification
Motor	400 V, 3 ph, 32 A fuse, shielded cable
Encoder	Shielded twisted pair, max. 30 m
Accelerometer	IEPE, 4 mA, BNC to gateway
PT100	3-wire, to gateway Modbus module
Earth	Separate PE to spindle housing, 10 mm²

3.6 Commissioning checklist

#	Check	OK
1	Mounting screws tightened to 49 Nm	
2	Pulley clamping set at 12 Nm	
3	Pulley alignment within limits (3.4)	
4	Belt tension 142 Hz	
5	Cooling flow 3–5 l/min, no leaks	
6	Sealing air 4–6 bar	
7	Rotary union tight	
8	Drawbar clamping force 18 kN	
9	All sensors read plausible values	
10	Encoder direction correct	
11	Emergency stop brakes within 3 s	
12	Guard interlock works	
13	Warm-up program completed	
14	Run-out at taper below 2 µm	
15	Vibration baseline recorded (3.7)	
16	Temperatures stable after 2 h	
17	No unusual noise	
18	Documentation handed over	
19	Spare parts on site (9.1)	
20	Operators trained	

3.7 Recording the vibration baseline

The baseline is the reference for all later trend checks. Record it after 2 hours of running at rated speed, with a typical tool and load.

1. Run the spindle at 12,000 rpm for 2 hours.
2. Record RMS, peak, kurtosis, bearing temperature, current and power as 10-minute averages.
3. Store the values in the maintenance system and on the machine record card.
4. Repeat after every bearing or belt replacement.

3.8 Drive parameters

Factory settings of the spindle drive. Change parameters only after consulting the machine builder; record every change in the maintenance log (10.2).

Parameter	Name	Value	Unit
P1000	Rated motor power	11.0	kW
P1005	Rated motor voltage	400	V
P1010	Rated motor current	21.0	A
P1015	Rated motor speed	7,500	rpm
P1020	Motor pole pairs	2	-
P1025	Max. motor speed	9,375	rpm
P1030	Belt ratio numerator	16	-
P1035	Belt ratio denominator	10	-
P1040	Max. spindle speed	15,000	rpm
P1045	Ramp-up time 0-12,000 rpm	4.0	s
P1050	Ramp-down time	3.0	s
P1055	Current limit	150	%
P1060	Speed controller gain	0.35	-
P1065	Speed controller integral time	18	ms
P1070	Current controller gain	12.5	-
P1075	Encoder lines	1,024	-
P1080	Encoder type	sin/cos	-
P1085	Speed window at speed	2	%
P1090	Zero speed threshold	20	rpm
P1095	Thermal motor model	on	-
P1100	Motor thermal time constant	25	min
P1105	Thermistor trip	140	degC
P1110	Overload warning	130	%
P1115	Brake resistor	33	Ohm
P1120	DC link warning	720	V
P1125	Switching frequency	8	kHz
P1130	Safe torque off	enabled	-
P1135	Safe stop 1 delay	3.0	s
P1140	Warm-up program	auto	-
P1145	Reduced-speed limit	70	%
P1150	Belt slip warning	2	%
P1155	Belt slip alarm	5	%

Notes

4 Operation

4.1 Start-up and warm-up

Cold bearings have a higher preload and poorly distributed grease. The warm-up program raises the speed in steps. It runs automatically when the spindle has been stopped longer than shown below.

Stopped for	Warm-up program	Duration
< 30 min	None	–
30 min – 4 h	50% for 5 min	5 min
4 h – 2 days	30% / 60% / 90% for 3 min each	9 min
> 2 days or after service	20% / 40% / 60% / 80% / 100% for 3 min each	15 min

4.2 Normal operating values

Speed	Vibration RMS	Rear bearing	Power (cutting)
6,000 rpm	0.9–1.3 mm/s	45–50 °C	5–7 kW
9,000 rpm	1.4–1.9 mm/s	50–55 °C	7–9 kW
12,000 rpm	1.8–2.4 mm/s	55–60 °C	10.5–11.5 kW
15,000 rpm	2.4–3.0 mm/s	60–66 °C	11–12 kW

Values vary with tool, material and ambient temperature. Compare each spindle with its own baseline (3.7) rather than with another spindle.

4.3 Speed and load limits

Limit	Value	Note
Continuous speed	12,000 rpm	Rated value
Short-time speed	15,000 rpm	Max. 20% of operating time
Continuous power	11 kW	S1
Peak power	15 kW for 60 s	S6
Max. axial tool force	3.5 kN	Drilling

4.4 Reduced-speed operation

If a bearing defect is suspected and the spindle must keep running until a planned stop, reduce the speed to 70% or less. Bearing load, heat and the energy of each impact fall with speed, which slows the damage and extends the safe run time. Check vibration and bearing temperature at least every hour and stop immediately if the RMS value exceeds 4.5 mm/s. Reduced speed lowers output; agree the plan with production.

4.5 Shutdown

1. Finish the current cycle.
2. Let the spindle run 2 minutes at 30% to cool down evenly.
3. Stop the spindle; keep sealing air on while coolant is present.
4. For stops longer than 2 days, switch off cooling and blow out the rotary union.

4.6 HMI messages

Code	Message	Meaning / action
H100	Spindle ready	Drive enabled, no fault.
H101	Warm-up running	Automatic warm-up program active (4.1).
H102	Warm-up skipped	Operator skipped warm-up; logged for maintenance.
H103	Speed limited	Speed override active, e.g. reduced-speed operation (4.4).
H104	Lubrication due	Re-greasing interval reached (5.3).
H105	Service due	Semi-annual service due (7.3).
H106	Belt tension check	Belt run-in period after replacement (7.7).
H107	Vibration high (warning band)	RMS above 4.0 mm/s; check trend (6.4).
H108	Vibration alarm	RMS above 4.5 mm/s for 30 min (6.7); plan a stop.
H109	Bearing temperature high	Rear bearing above 75 °C; check lubrication and load.
H110	Bearing temperature trip	Rear bearing above 85 °C; spindle stopped.
H111	Motor temperature high	Winding above 120 °C; check cooling (7.8).
H112	Coolant flow low	Spindle cooling flow below 3 l/min (7.8).
H113	Encoder signal fault	Speed feedback missing or noisy (7.9).
H114	Drive overcurrent	Current above 150% for 2 s; check tool and bearings.
H115	Drawbar not clamped	Tool clamping pressure below limit.
H116	Sensor value out of range	A sensor reports a value outside its range (6.8).
H117	Sensor signal stale	No update for more than 5 s (6.8).
H118	Door open	Machine guard open; spindle stays in safe stop.
H119	Emergency stop	E-stop pressed; reset after the cause is cleared.
H120	Air pressure low	Sealing air below 4 bar.
H121	Chiller fault	Cooling unit reports a fault (7.8).
H122	Belt slip detected	Motor and spindle speed differ by more than 2% (7.7).
H123	Maintenance mode	Spindle under lockout; only jog at 10% allowed.

4.7 Cutting data for drilling

Starting values for solid carbide drills with internal coolant. Spindle speed is limited to 15,000 rpm. Reduce the values by 20% for unstable clamping or long tools.

Aluminium cast AlSi10Mg

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
4 mm	188 m/min	15,000 rpm	0.114 mm/rev	1,708 mm/min
5 mm	236 m/min	15,000 rpm	0.127 mm/rev	1,909 mm/min
6 mm	283 m/min	15,000 rpm	0.139 mm/rev	2,091 mm/min
8 mm	300 m/min	11,937 rpm	0.161 mm/rev	1,922 mm/min
10 mm	300 m/min	9,549 rpm	0.180 mm/rev	1,719 mm/min
12 mm	300 m/min	7,958 rpm	0.197 mm/rev	1,569 mm/min
14 mm	300 m/min	6,821 rpm	0.213 mm/rev	1,453 mm/min
16 mm	300 m/min	5,968 rpm	0.228 mm/rev	1,359 mm/min
20 mm	300 m/min	4,775 rpm	0.255 mm/rev	1,215 mm/min

Aluminium wrought 6082

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
4 mm	188 m/min	15,000 rpm	0.126 mm/rev	1,897 mm/min
5 mm	236 m/min	15,000 rpm	0.141 mm/rev	2,121 mm/min
6 mm	283 m/min	15,000 rpm	0.155 mm/rev	2,324 mm/min
8 mm	350 m/min	13,926 rpm	0.179 mm/rev	2,491 mm/min
10 mm	350 m/min	11,141 rpm	0.200 mm/rev	2,228 mm/min
12 mm	350 m/min	9,284 rpm	0.219 mm/rev	2,034 mm/min
14 mm	350 m/min	7,958 rpm	0.237 mm/rev	1,883 mm/min
16 mm	350 m/min	6,963 rpm	0.253 mm/rev	1,762 mm/min
20 mm	350 m/min	5,570 rpm	0.283 mm/rev	1,576 mm/min

Grey cast iron EN-GJL-250

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
4 mm	110 m/min	8,754 rpm	0.139 mm/rev	1,218 mm/min
5 mm	110 m/min	7,003 rpm	0.156 mm/rev	1,089 mm/min
6 mm	110 m/min	5,836 rpm	0.170 mm/rev	994 mm/min
8 mm	110 m/min	4,377 rpm	0.197 mm/rev	861 mm/min
10 mm	110 m/min	3,501 rpm	0.220 mm/rev	770 mm/min
12 mm	110 m/min	2,918 rpm	0.241 mm/rev	703 mm/min
14 mm	110 m/min	2,501 rpm	0.260 mm/rev	651 mm/min
16 mm	110 m/min	2,188 rpm	0.278 mm/rev	609 mm/min
20 mm	110 m/min	1,751 rpm	0.311 mm/rev	545 mm/min

Steel C45

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
4 mm	90 m/min	7,162 rpm	0.101 mm/rev	725 mm/min
5 mm	90 m/min	5,730 rpm	0.113 mm/rev	648 mm/min
6 mm	90 m/min	4,775 rpm	0.124 mm/rev	592 mm/min
8 mm	90 m/min	3,581 rpm	0.143 mm/rev	512 mm/min
10 mm	90 m/min	2,865 rpm	0.160 mm/rev	458 mm/min
12 mm	90 m/min	2,387 rpm	0.175 mm/rev	418 mm/min
14 mm	90 m/min	2,046 rpm	0.189 mm/rev	387 mm/min
16 mm	90 m/min	1,790 rpm	0.202 mm/rev	362 mm/min
20 mm	90 m/min	1,432 rpm	0.226 mm/rev	324 mm/min

Stainless steel 1.4301

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
4 mm	60 m/min	4,775 rpm	0.063 mm/rev	302 mm/min
5 mm	60 m/min	3,820 rpm	0.071 mm/rev	270 mm/min
6 mm	60 m/min	3,183 rpm	0.077 mm/rev	247 mm/min
8 mm	60 m/min	2,387 rpm	0.089 mm/rev	214 mm/min
10 mm	60 m/min	1,910 rpm	0.100 mm/rev	191 mm/min
12 mm	60 m/min	1,592 rpm	0.110 mm/rev	174 mm/min
14 mm	60 m/min	1,364 rpm	0.118 mm/rev	161 mm/min
16 mm	60 m/min	1,194 rpm	0.126 mm/rev	151 mm/min
20 mm	60 m/min	955 rpm	0.141 mm/rev	135 mm/min

Brass CuZn39Pb3

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
4 mm	160 m/min	12,732 rpm	0.114 mm/rev	1,449 mm/min
5 mm	160 m/min	10,186 rpm	0.127 mm/rev	1,296 mm/min
6 mm	160 m/min	8,488 rpm	0.139 mm/rev	1,183 mm/min
8 mm	160 m/min	6,366 rpm	0.161 mm/rev	1,025 mm/min
10 mm	160 m/min	5,093 rpm	0.180 mm/rev	917 mm/min
12 mm	160 m/min	4,244 rpm	0.197 mm/rev	837 mm/min
14 mm	160 m/min	3,638 rpm	0.213 mm/rev	775 mm/min
16 mm	160 m/min	3,183 rpm	0.228 mm/rev	725 mm/min
20 mm	160 m/min	2,546 rpm	0.255 mm/rev	648 mm/min

4.8 Cutting data for milling

Starting values for three-flute carbide end mills, side milling at 0.5 x D radial depth.

Aluminium cast AlSi10Mg

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
6 mm	283 m/min	15,000 rpm	0.049 mm/tooth	2,196 mm/min
8 mm	345 m/min	13,727 rpm	0.056 mm/tooth	2,321 mm/min
10 mm	345 m/min	10,982 rpm	0.063 mm/tooth	2,076 mm/min
12 mm	345 m/min	9,151 rpm	0.069 mm/tooth	1,895 mm/min
16 mm	345 m/min	6,864 rpm	0.080 mm/tooth	1,641 mm/min
20 mm	345 m/min	5,491 rpm	0.089 mm/tooth	1,468 mm/min
25 mm	345 m/min	4,393 rpm	0.100 mm/tooth	1,313 mm/min

Aluminium wrought 6082

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
6 mm	283 m/min	15,000 rpm	0.054 mm/tooth	2,440 mm/min
8 mm	377 m/min	15,000 rpm	0.063 mm/tooth	2,817 mm/min
10 mm	402 m/min	12,812 rpm	0.070 mm/tooth	2,691 mm/min
12 mm	402 m/min	10,677 rpm	0.077 mm/tooth	2,456 mm/min
16 mm	402 m/min	8,007 rpm	0.089 mm/tooth	2,127 mm/min
20 mm	402 m/min	6,406 rpm	0.099 mm/tooth	1,902 mm/min
25 mm	402 m/min	5,125 rpm	0.111 mm/tooth	1,702 mm/min

Grey cast iron EN-GJL-250

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
6 mm	126 m/min	6,711 rpm	0.060 mm/tooth	1,201 mm/min
8 mm	126 m/min	5,033 rpm	0.069 mm/tooth	1,040 mm/min
10 mm	126 m/min	4,027 rpm	0.077 mm/tooth	930 mm/min
12 mm	126 m/min	3,356 rpm	0.084 mm/tooth	849 mm/min
16 mm	126 m/min	2,517 rpm	0.097 mm/tooth	735 mm/min
20 mm	126 m/min	2,013 rpm	0.109 mm/tooth	658 mm/min
25 mm	126 m/min	1,611 rpm	0.122 mm/tooth	588 mm/min

Steel C45

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
6 mm	103 m/min	5,491 rpm	0.043 mm/tooth	715 mm/min
8 mm	103 m/min	4,118 rpm	0.050 mm/tooth	619 mm/min
10 mm	103 m/min	3,295 rpm	0.056 mm/tooth	553 mm/min
12 mm	103 m/min	2,745 rpm	0.061 mm/tooth	505 mm/min
16 mm	103 m/min	2,059 rpm	0.071 mm/tooth	438 mm/min

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
20 mm	103 m/min	1,647 rpm	0.079 mm/tooth	391 mm/min
25 mm	103 m/min	1,318 rpm	0.089 mm/tooth	350 mm/min

Stainless steel 1.4301

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
6 mm	69 m/min	3,661 rpm	0.027 mm/tooth	298 mm/min
8 mm	69 m/min	2,745 rpm	0.031 mm/tooth	258 mm/min
10 mm	69 m/min	2,196 rpm	0.035 mm/tooth	231 mm/min
12 mm	69 m/min	1,830 rpm	0.038 mm/tooth	211 mm/min
16 mm	69 m/min	1,373 rpm	0.044 mm/tooth	182 mm/min
20 mm	69 m/min	1,098 rpm	0.049 mm/tooth	163 mm/min
25 mm	69 m/min	879 rpm	0.055 mm/tooth	146 mm/min

Brass CuZn39Pb3

Tool diameter	Cutting speed	Spindle speed	Feed	Feed rate
6 mm	184 m/min	9,762 rpm	0.049 mm/tooth	1,429 mm/min
8 mm	184 m/min	7,321 rpm	0.056 mm/tooth	1,238 mm/min
10 mm	184 m/min	5,857 rpm	0.063 mm/tooth	1,107 mm/min
12 mm	184 m/min	4,881 rpm	0.069 mm/tooth	1,011 mm/min
16 mm	184 m/min	3,661 rpm	0.080 mm/tooth	875 mm/min
20 mm	184 m/min	2,928 rpm	0.089 mm/tooth	783 mm/min
25 mm	184 m/min	2,343 rpm	0.100 mm/tooth	700 mm/min

4.9 Tool holders and balancing

Unbalanced tool holders are a common cause of high vibration that is wrongly blamed on the spindle. Balance tool assemblies to G2.5 at the operating speed; for finishing above 12,000 rpm use G1.0.

Spindle speed	G2.5: residual unbalance	G1.0: residual unbalance	Max. for HSK-A63 holder (0.8 kg)
3,000 rpm	7.96 g·mm/kg	3.18 g·mm/kg	6.37 g·mm
4,000 rpm	5.97 g·mm/kg	2.39 g·mm/kg	4.77 g·mm
5,000 rpm	4.77 g·mm/kg	1.91 g·mm/kg	3.82 g·mm
6,000 rpm	3.98 g·mm/kg	1.59 g·mm/kg	3.18 g·mm
7,000 rpm	3.41 g·mm/kg	1.36 g·mm/kg	2.73 g·mm
8,000 rpm	2.98 g·mm/kg	1.19 g·mm/kg	2.39 g·mm
9,000 rpm	2.65 g·mm/kg	1.06 g·mm/kg	2.12 g·mm
10,000 rpm	2.39 g·mm/kg	0.95 g·mm/kg	1.91 g·mm
11,000 rpm	2.17 g·mm/kg	0.87 g·mm/kg	1.74 g·mm
12,000 rpm	1.99 g·mm/kg	0.80 g·mm/kg	1.59 g·mm
13,000 rpm	1.84 g·mm/kg	0.73 g·mm/kg	1.47 g·mm
14,000 rpm	1.71 g·mm/kg	0.68 g·mm/kg	1.36 g·mm
15,000 rpm	1.59 g·mm/kg	0.64 g·mm/kg	1.27 g·mm

5 Lubrication

5.1 Lubrication system

The front bearings are greased for life. The rear bearing (6205-2RS) is re-greased through the nipple on the rear cover. Using the wrong grease, too much grease or too little are the most common causes of rear bearing damage.

5.2 Grease specification

Property	Requirement
Type	Synthetic ester oil, lithium-complex thickener
NLGI class	2
Base oil viscosity at 40 °C	22–25 mm²/s
Speed factor	> 1,000,000
Product	GRS-HT2 or approved equivalent

5.3 Intervals and quantities

Operating speed	Interval	Quantity (rear bearing)
up to 9,000 rpm	3,000 h	1.5 g
9,000–12,000 rpm	2,000 h	1.5 g
above 12,000 rpm	1,200 h	1.0 g

5.4 Re-greasing procedure

1. Run the spindle until warm (bearing above 45 °C).
2. Stop and lock out (1.4).
3. Clean the nipple.
4. Inject the quantity from 5.3 with the metering gun; never more.
5. Run 10 min at 30%, 10 min at 60%, then normal operation.
6. Record date, operating hours and quantity (5.6).

5.5 Over-greasing and run-in

After re-greasing the bearing temperature can rise by 5–10 °C for a few hours while the grease spreads. This is normal and settles by itself. If the temperature keeps rising or exceeds 75 °C, stop and check for over-greasing.

5.6 Lubrication records

Date	Operating hours	Quantity	Name	Remarks

5.7 Approved lubricants

Use	Lubricant	Note
Rear bearing	GRS-HT2 (ester / Li-complex)	Standard
Rear bearing, high ambient	GRS-HT3 (ester / polyurea)	Above 35 °C ambient
Front bearings (factory fill)	GRS-SP1	Do not re-grease
Belt drive	None	Never lubricate belts
Drawbar	Molybdenum paste MP-2	Thin film at service
O-rings	Silicone grease SG-1	Assembly only

6 Condition monitoring

6.1 Signals and what they show

Signal	What it shows	Typical cause of a change
Vibration RMS	Overall vibration energy	Unbalance, misalignment, late bearing damage
Vibration peak	Strongest single impact	Rolling-element damage, chatter
Kurtosis	How spiky the vibration is (3 = normal)	Early bearing damage (impacts)
Bearing temperature	Friction and lubrication	Grease, preload, damage
Motor current	Load on the drive	Cutting load, friction
Power	Energy taken	Friction from worn bearings, blunt tools
Speed	Actual spindle speed	Belt slip, drive problems

6.2 Normal ranges

Normal values at 12,000 rpm are listed in 2.3. A healthy spindle stays within ±10% of its own baseline for RMS and power, and within ±3 °C for bearing temperature.

6.3 Vibration severity zones

Zone	RMS (mm/s)	Meaning
A	< 1.4	New machine, very good
B	1.4–2.8	Unrestricted long-term operation
C	2.8–4.5	Restricted operation; plan action
D	> 4.5	Damage likely; stop as soon as possible

Zones follow the plant standard PS-VIB-01, which is based on ISO 10816-3 for machines of this size. The classic alarm (4.5 mm/s for 30 min) marks the start of zone D.

6.4 Reading trends

A single value says little; the trend says a lot. Look at the last 24 hours and the last 7 days and ask: is it rising, how fast, and are other signals moving with it?

- Slow rise over weeks: normal wear, plan the next service.
- Rise over days, with kurtosis and temperature: bearing damage, see 7.4.
- Rise over days with normal kurtosis: misalignment or looseness, see 7.6.
- Sudden jump after a tool change: unbalanced tool holder, check the tool.
- Rise only at one speed: resonance, check mounting and speed program.

6.5 Kurtosis and peak values

Kurtosis measures how impulsive the vibration is. A healthy bearing gives values around 3. Small defects on a raceway produce short impacts that lift kurtosis above 4 long before the overall RMS value reaches the alarm limit. Kurtosis can fall again late in the damage, when the defect spreads and the signal becomes noisy, so always read it together with RMS and temperature.

6.6 Temperature, current and power

A worn rear bearing creates friction: bearing temperature rises a few degrees and the drive draws 5–10% more power at the same cutting load. Power also rises with blunt tools, so compare at the same program and tool.

6.7 Alarm limits

Signal	Warning	Alarm / trip
Vibration RMS	4.0 mm/s	4.5 mm/s for 30 min
Kurtosis	4.0 for 1 h	–
Rear bearing temperature	75 °C	85 °C (trip)
Motor winding	120 °C	140 °C (trip)
Current	130% for 10 s	150% for 2 s (trip)

6.8 Data quality

Problem	How it shows	Action
Out of range	Value outside sensor range	Check sensor, cable
Stale	No update for more than 5 s	Check gateway and network
Drift	Slow change with no change in other signals	Recalibrate (7.9)
Spikes	Single extreme values	Check shielding and earth

6.9 Predictive warnings

Plants that run a predictive model on these signals receive a breakdown risk for the next 72 hours. The model uses the same signals as this chapter plus the operating hours since the last service. Treat a high risk like a vibration warning: check the trend, follow section 7.4 or 7.6, and plan the repair at the next possible stop.

6.10 Bearing defect frequencies

Use these frequencies with a handheld analyser or the envelope spectrum of the gateway. A peak at BPFO and its harmonics with raised kurtosis means outer-race damage, the most common fault of the rear bearing.

Rear bearing 6205-2RS - BPFO 3.572x, BPFI 5.428x, BSF 2.321x, FTF 0.397x running speed

Spindle speed	Shaft 1x	BPFO (outer race)	BPFI (inner race)	BSF (ball)	FTF (cage)
3,000 rpm	50.0 Hz	178.6 Hz	271.4 Hz	116.1 Hz	19.8 Hz
3,750 rpm	62.5 Hz	223.2 Hz	339.3 Hz	145.1 Hz	24.8 Hz
4,500 rpm	75.0 Hz	267.9 Hz	407.1 Hz	174.1 Hz	29.8 Hz
5,250 rpm	87.5 Hz	312.5 Hz	475.0 Hz	203.1 Hz	34.7 Hz
6,000 rpm	100.0 Hz	357.2 Hz	542.8 Hz	232.1 Hz	39.7 Hz
6,750 rpm	112.5 Hz	401.8 Hz	610.7 Hz	261.1 Hz	44.6 Hz
7,500 rpm	125.0 Hz	446.5 Hz	678.5 Hz	290.2 Hz	49.6 Hz
8,250 rpm	137.5 Hz	491.1 Hz	746.4 Hz	319.2 Hz	54.6 Hz
9,000 rpm	150.0 Hz	535.8 Hz	814.2 Hz	348.2 Hz	59.5 Hz
9,750 rpm	162.5 Hz	580.4 Hz	882.1 Hz	377.2 Hz	64.5 Hz
10,500 rpm	175.0 Hz	625.1 Hz	949.9 Hz	406.2 Hz	69.5 Hz
11,250 rpm	187.5 Hz	669.7 Hz	1017.8 Hz	435.2 Hz	74.4 Hz
12,000 rpm	200.0 Hz	714.4 Hz	1085.6 Hz	464.3 Hz	79.4 Hz
12,750 rpm	212.5 Hz	759.0 Hz	1153.5 Hz	493.3 Hz	84.3 Hz
13,500 rpm	225.0 Hz	803.7 Hz	1221.3 Hz	522.3 Hz	89.3 Hz
14,250 rpm	237.5 Hz	848.3 Hz	1289.2 Hz	551.3 Hz	94.3 Hz
15,000 rpm	250.0 Hz	893.0 Hz	1357.0 Hz	580.3 Hz	99.2 Hz

Front bearing 7008 (each) - BPFO 6.542x, BPFI 8.458x, BSF 3.720x, FTF 0.436x running speed

Spindle speed	Shaft 1x	BPFO (outer race)	BPFI (inner race)	BSF (ball)	FTF (cage)
3,000 rpm	50.0 Hz	327.1 Hz	422.9 Hz	186.0 Hz	21.8 Hz
3,750 rpm	62.5 Hz	408.9 Hz	528.6 Hz	232.5 Hz	27.3 Hz
4,500 rpm	75.0 Hz	490.7 Hz	634.3 Hz	279.0 Hz	32.7 Hz
5,250 rpm	87.5 Hz	572.4 Hz	740.1 Hz	325.5 Hz	38.2 Hz
6,000 rpm	100.0 Hz	654.2 Hz	845.8 Hz	372.0 Hz	43.6 Hz
6,750 rpm	112.5 Hz	736.0 Hz	951.5 Hz	418.5 Hz	49.1 Hz
7,500 rpm	125.0 Hz	817.8 Hz	1057.2 Hz	465.0 Hz	54.5 Hz
8,250 rpm	137.5 Hz	899.5 Hz	1163.0 Hz	511.5 Hz	60.0 Hz
9,000 rpm	150.0 Hz	981.3 Hz	1268.7 Hz	558.0 Hz	65.4 Hz
9,750 rpm	162.5 Hz	1063.1 Hz	1374.4 Hz	604.5 Hz	70.9 Hz
10,500 rpm	175.0 Hz	1144.9 Hz	1480.1 Hz	651.0 Hz	76.3 Hz
11,250 rpm	187.5 Hz	1226.6 Hz	1585.9 Hz	697.5 Hz	81.8 Hz

Spindle speed	Shaft 1x	BPFO (outer race)	BPFI (inner race)	BSF (ball)	FTF (cage)
12,000 rpm	200.0 Hz	1308.4 Hz	1691.6 Hz	744.0 Hz	87.2 Hz
12,750 rpm	212.5 Hz	1390.2 Hz	1797.3 Hz	790.5 Hz	92.7 Hz
13,500 rpm	225.0 Hz	1472.0 Hz	1903.0 Hz	837.0 Hz	98.1 Hz
14,250 rpm	237.5 Hz	1553.8 Hz	2008.7 Hz	883.5 Hz	103.6 Hz
15,000 rpm	250.0 Hz	1635.5 Hz	2114.5 Hz	930.0 Hz	109.0 Hz

6.11 Example trend patterns

The examples below come from typical cases. They show how the signals move together for different faults.

Example 1 - early rear bearing damage

RMS 2.1 to 2.5 mm/s over five days, kurtosis 3.0 to 4.2, rear bearing +3 °C, power +3%. Interpretation: stage 1 bearing damage. Action: plan replacement within 1-2 weeks, re-check daily.

Example 2 - fast bearing deterioration

RMS 3.2 to 4.3 mm/s within 6 hours, kurtosis above 5, rear bearing +8 °C, power +8%. Interpretation: stage 2 to 3 damage; seizure possible within days. Action: section 7.4 immediately; reduce speed (4.4) until the planned stop.

Example 3 - pulley misalignment after belt change

RMS 1.9 to 3.4 mm/s over three days, kurtosis 3.1, strongest peak at 2x running speed, bearing +2 °C. Action: section 7.6.

Example 4 - cooling problem

Rear bearing 58 to 79 °C within 10 hours, front bearings also warmer, vibration unchanged, coolant flow 2.1 l/min. Action: section 7.8, replace filter.

Example 5 - unbalanced tool holder

RMS jumps from 2.0 to 3.6 mm/s at a tool change and returns to 2.0 with the next tool. Action: balance or replace the tool holder (4.8).

Example 6 - sensor problem

RMS falls from 2.1 to 0.6 mm/s with no change in power or temperature. Action: check accelerometer mounting and cable (7.9).

Example 7 - load change

RMS +20% and power +6% for four hours during a heavy roughing program, then back to normal. Interpretation: process load, no damage. Action: none; compare at the same program.

Example 8 - reduced speed

After switching to 70% speed, RMS falls by about one third and bearing temperature by 4-6 °C. Interpretation: expected; a bearing defect is still present.

6.12 Measuring with a handheld analyser

1. Measure at the same point as the fixed sensor (rear housing, radial).
2. Use a magnet base on a clean, flat surface.
3. Measure at 12,000 rpm with the warm-up finished.
4. Record RMS (10 Hz-1 kHz), peak, kurtosis and an envelope spectrum.
5. Compare with the baseline and the previous measurement.

7 Maintenance

7.1 Daily checks

#	Check	Normal
1	Listen for unusual noise	Even, no knocking or squealing
2	Vibration and temperature on HMI	Inside the ranges in 2.3
3	Coolant leaks at spindle nose	Dry
4	Sealing air pressure	4–6 bar
5	Chip build-up at belt cover	Clean
6	Tool clamping message	Clamped
7	Active messages	None

7.2 Weekly checks

#	Check	Normal
1	Clean the motor fan cover	Free airflow
2	Belt cover tight	No rattling
3	Coolant filter differential pressure	< 0.8 bar
4	Trend of RMS and temperature, last 7 days	No rise
5	Drawbar clamping force (test gauge)	16–20 kN
6	Cable entries and connectors	Tight, undamaged

7.3 Semi-annual service

Every 4,000 operating hours or 6 months, whichever comes first. Allow 1 hour. Record all measured values on the form at the end of this section; they are the reference for the next service.

#	Work step	Limit / value
1	Record RMS, peak, kurtosis, temperatures at 12,000 rpm	Compare with baseline
2	Lock out (1.4)	–
3	Remove belt cover; inspect belt teeth and back	No cracks, no missing teeth
4	Measure belt tension	128–136 Hz
5	Check pulley alignment with laser tool	Limits in 3.4
6	Re-grease rear bearing (5.4)	1.5 g
7	Turn shaft by hand	Smooth, no notching
8	Check radial play at the nose	< 2 µm
9	Check drawbar clamping force	16–20 kN
10	Clean and inspect taper	No fretting
11	Replace coolant filter FLT-SPC-10	–
12	Check coolant flow	3–5 l/min
13	Check sealing air	4–6 bar
14	Check sensor mounting torque (accelerometer)	8 Nm

#	Work step	Limit / value
15	Check PT100 reading against hand thermometer	± 1 °C
16	Inspect cables and connectors	Undamaged
17	Clean motor fan and fins	–
18	Refit belt cover	–
19	Run warm-up program (4.1)	–
20	Record RMS, peak, kurtosis, temperatures after service	Within baseline ±10%
21	Reset service counter on HMI	–
22	Enter the service in the maintenance system	–

Service record

Value	Before service	After service	Baseline
Vibration RMS (mm/s)			
Vibration peak (mm/s)			
Kurtosis			
Rear bearing temperature (°C)			
Motor current (A)			
Power (kW)			
Belt tension (Hz)			
Clamping force (kN)			
Operating hours			

7.4 Bearing wear

If vibration RMS rises more than 30% within 24 h together with kurtosis above 4, inspect the rear bearing (6205-2RS). Continued operation above 4.5 mm/s may cause seizure within 48–96 h. Reducing spindle speed to 70% lowers bearing load and extends safe run time.

Bearing wear develops in stages. In the first stage, small defects on a raceway create short impacts: kurtosis and the peak value rise while the overall RMS value hardly changes. In the second stage the defect grows; RMS, bearing temperature and power rise together, often by 30–50% within a day. In the last stage RMS climbs above the 4.5 mm/s alarm limit and the bearing can seize. Machining quality suffers early: bores drift out of tolerance because the shaft wobbles.

Sign	Bearing wear	Misalignment (7.6)
Kurtosis	Above 4	Normal (about 3)
Vibration RMS	Rising, then rising fast	Rising steadily
Strongest frequency	Bearing defect frequencies	2x running speed
Bearing temperature	Rising	Slightly rising
Power	+5–10%	+3–5%

Action: confirm the trend over the last 24 hours, plan the replacement of the rear bearing (7.5) at the next possible stop, and order the part (SKF-6205) if it is not in stock. Until the stop, run at reduced speed (4.4) and check vibration every hour. A technician with certificate spindle-200 must do the replacement.

7.5 Rear bearing replacement

Downtime about 2.5 hours with parts and tools prepared. Only technicians with certificate spindle-200.

1. Lock out (1.4) and let the spindle cool below 40 °C.
2. Remove the belt cover and release the belt tension; take off the belt.
3. Remove the spindle pulley (clamping set).
4. Remove the rear cover and the PT100 probe.
5. Pull the rear bearing with the extractor; do not hit the shaft.
6. Inspect the shaft seat and the housing bore for wear and fretting.
7. Clean the seat; check the shaft diameter (25 k5).
8. Heat the new bearing (SKF-6205) to 80 °C with the induction heater.
9. Slide it on until it touches the shoulder; hold 30 s.
10. Let it cool; fill 1.5 g of grease GRS-HT2.
11. Refit rear cover, PT100 and pulley.
12. Fit the belt; set tension to 142 Hz (new belt) or 132 Hz (used belt).
13. Align the pulleys (3.4).
14. Turn by hand: smooth, no noise.
15. Run the warm-up program (4.1).
16. Record a new vibration baseline (3.7).
17. Enter the work in the maintenance system with findings.

Tools and parts for 7.5

Item	Specification
Bearing	SKF-6205 (6205-2RS), 1 piece
Grease	GRS-HT2, 1.5 g
Extractor	Two-arm puller, 25 mm
Induction heater	Set to 80 °C
Frequency meter	Belt tension
Laser pulley tool	Alignment (3.4)
Torque wrench	5–60 Nm
Thermometer	Contact, ±1 °C

Screw	Torque
Rear cover M6	10 Nm
Pulley clamping set	12 Nm
Belt cover M5	6 Nm
PT100 gland	5 Nm

Note Always inspect the old bearing. Write in the work order what you found (for example outer race worn, cage damage, discoloured grease). These findings help the predictive model learn.

7.6 Misalignment

Misalignment shows as high vibration at 2× running speed with normal kurtosis.

Pulley misalignment and loose mounting both raise the vibration level steadily over days. Unlike bearing damage, kurtosis stays near 3 and the bearing temperature changes little. Left alone, misalignment overloads the rear bearing and can lead to bearing damage, so both can be present at the same time.

1. Check the trend: RMS rising, kurtosis normal?
2. Lock out (1.4) and remove the belt cover.
3. Check pulley alignment with the laser tool (limits in 3.4).
4. Check the mounting screws (49 Nm).
5. Correct the alignment, set the belt tension.
6. Run and compare RMS with the baseline.

7.7 Drive belt replacement

Replace the belt every 8,000 hours or when teeth are cracked or missing. A worn belt causes speed deviations and slip messages (H122).

1. Lock out (1.4); remove the belt cover.
2. Release the motor tensioning screws; remove the old belt.
3. Check both pulleys for worn teeth.
4. Fit the new belt BLT-HTD-8M without levering it over the flanges.
5. Set tension to 142 Hz; align the pulleys (3.4).
6. After 24 h of running, re-tension to 132 Hz.

7.8 Spindle cooling

The cooling circuit removes heat from the front bearings and the motor. Low flow or a blocked filter makes temperatures rise within hours while vibration stays normal. Check the filter (FLT-SPC-10), the flow (3–5 l/min) and the chiller.

7.9 Encoder and sensors

Check the accelerometer mounting torque (8 Nm) and the cable at every service. A loose sensor shows lower vibration than real. Recalibrate the PT100 against a contact thermometer once a year.

8 Troubleshooting

8.1 Symptoms

Symptom	Possible cause	Action
RMS rising, kurtosis above 4, bearing warm	Rear bearing wear	7.4, 7.5
RMS rising, kurtosis normal	Pulley misalignment, loose mounting	7.6
High temperature, vibration normal	Cooling, over-greasing, preload	5.5, 7.8
Bores out of tolerance	Bearing play, tool wear	7.4; check tool
Speed deviation, slip message	Belt worn or loose	7.7
Vibration high after tool change	Unbalanced tool holder	Check tool
Noise like grinding	Late bearing damage	Stop; 7.5
Coolant at the nose	Seal or rotary union	Replace seal
Power up 5–10% at same load	Friction from worn bearing	7.4
Current spikes	Tool crash, chips in the drive	Inspect

8.2 Alarm list

Alarm	Text	Likely cause	Section
SP-101	Vibration RMS warning	Unbalance, worn tool, early bearing damage	6.4, 7.4
SP-102	Vibration RMS alarm	Bearing damage, misalignment, loose mounting	7.4, 7.6
SP-103	Vibration peak high	Impacts from rolling-element damage or chatter	6.5
SP-104	Kurtosis high	Rolling-element bearing damage	7.4
SP-110	Rear bearing temperature warning	Lubrication, preload, load	5.5, 7.4
SP-111	Rear bearing temperature trip	Bearing failure, no lubrication	7.5
SP-112	Front bearing temperature warning	Preload too high, cooling	7.8
SP-120	Motor winding temperature warning	Blocked fan, overload	7.8
SP-121	Motor winding temperature trip	Cooling failure, overload	7.8
SP-130	Coolant flow low	Filter blocked, pump fault	7.8
SP-131	Coolant temperature high	Chiller fault, fouled exchanger	7.8
SP-140	Belt slip	Belt worn or tension low	7.7
SP-141	Belt tension low	Belt stretch after run-in	7.7
SP-150	Encoder fault	Cable, connector, encoder	7.9
SP-151	Speed deviation	Belt slip, drive tuning	7.7, 7.9
SP-160	Drive overcurrent	Tool crash, seizure, short circuit	8.1
SP-161	Drive overvoltage	Braking too fast, supply	8.1

Alarm	Text	Likely cause	Section
SP-162	Drive undervoltage	Supply dip	8.1
SP-170	Drawbar unclamped	Clamping pressure, spring fatigue	8.1
SP-171	Tool not detected	Sensor, tool holder	8.1
SP-180	Sealing air low	Compressed air supply	8.1
SP-190	Sensor out of range	Sensor or cable fault	6.8
SP-191	Sensor stale	Gateway or network	6.8
SP-192	Sensor drift suspected	Calibration due	7.9
SP-199	Service overdue	Semi-annual service not done	7.3

9 Spare parts

9.1 Recommended spares on site

Part number	Description	Qty per spindle	Keep on site
SKF-6205	Rear bearing 6205-2RS	1	1 per spindle
SKF-7008	Front angular contact bearing	2	1 pair per site
BLT-HTD-8M	Drive belt HTD 8M	1	1 per spindle
FLT-SPC-10	Coolant filter	1	2
GRS-HT2	Bearing grease, 400 g	–	1
SEAL-LAB-40	Labyrinth seal kit	1	1
VIB-SENS-100	Accelerometer	1	1
PT100-PROBE	Temperature probe	1	1
ENC-1024	Encoder	1	1

9.2 Ordering

Order parts with the part number and the spindle serial number. Standard bearings are also available from bearing distributors; use only the exact type and clearance class.

10 Appendices

10.1 Tightening torques

Size	Steel 8.8	Steel 10.9
M5	6 Nm	8.5 Nm
M6	10 Nm	14 Nm
M8	25 Nm	35 Nm
M10	49 Nm	69 Nm
M12	86 Nm	120 Nm

10.2 Maintenance log

Date	Hours	Work done	Findings	Name

10.3 Glossary

Term	Meaning
RMS	Root mean square: overall vibration level
Kurtosis	How spiky a signal is; 3 = normal, above 4 = impacts
Baseline	The spindle's own normal values, recorded after installation or repair
Zone C / D	Vibration severity zones (6.3)
LOTO	Lockout and tagout (1.4)
RUL	Remaining useful life: estimated time until breakdown